PART 2 – WRITING A CLEAR SCIENTIFIC MESSAGE

- Principles of effective scientific writing
- Grammar rules
- Style rules

Dr. Carolina Borges Merjane
Scientific and Medical Writing and Training



Scientific writing follows the same core grammar rules as standard English, but:
it applies them with a strong emphasis on **clarity, precision, objectivity, and consistency**.

Can a knowledgeable reader understand exactly what I mean on the first reading, with **no ambiguity**?

The focus should be on **understanding the science being communicated, not the writing.**

There is a set of rules and guidelines that are conventionally used to achieve this goal.

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First steps and helpful resources:

- Successful examples – recent completed thesis and dissertations; published papers; grant proposals
- Scientific literature – published papers in the field (e.g. for jargon, field conventions, knowledge)
- Read guidelines (e.g. Thesis guidelines; journal guidelines)
- Reference Books
- Principles of the American Medical Association (AMA Manual of Style, latest edition)
- LLMs (appropriately)
- Write, write and…. Re-write.
- Get feedback from peers

WRITING A CLEAR SCIENTIFIC MESSAGE
Grammar rules

1. Voice: Active voice is preferred over Passive voice
2. Tenses: Simple tenses (present and past) are preferred over Perfect tense
3. Fewer Prepositions: Avoid excessive prepositional phrases
4. Fewer modifiers: Keep clear and concise
5. Punctuation: Clear marks, clear meaning

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1. Voice: Active Voice is preferred over Passive Voice

The English language has two "voices" :
grammatical feature that shows the relationship between the subject of a sentence and the action of the verb

The subject performs the action (**active voice**) or receives the action (**passive voice**).

Active voice: *We analyzed the data using linear regression.*

Passive voice: *The data were analyzed using linear regression.*

Active voice: *The researchers collected blood samples from 200 participants.*

Passive voice: *Blood samples were collected from 200 participants.*

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1. Voice: Active Voice is preferred over Passive Voice

- Active voice is preferred because it is **clearer, shorter, easier to read, and more transparent** about who did the action.
- Passive voice is avoided because it:
  - reduces clarity
  - makes sentences wordier
  - decreases readability → increases confusion
  - weakens accountability and transparency
- Use passive voice could be used when the action or recipient matters more than the actor (e.g. methods section)
- Note that certain journals will specifically state **they expect active voice** (e.g. Nature, BMJ).

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1. Voice: Active Voice is preferred over Passive Voice

Active Voice	Why It's Better	Passive Voice	Why It's Worse
We found that sleep deprivation impaired memory performance.	Clearly states who made the observation.	It was found that sleep deprivation impaired memory performance.	Wordy and hides the actor.
Our results suggest that oxidative stress contributes to neuronal degeneration.	Directly attributes the interpretation to the authors.	It is suggested that oxidative stress contributes to neuronal degeneration.	Sounds vague and impersonal.
We propose a model linking inflammation to cognitive decline.	Clearly communicates the authors' contribution.	A model linking inflammation to cognitive decline is proposed.	Less direct and less engaging.
We hypothesize that this pathway regulates synaptic plasticity.	Makes clear that this is the authors' hypothesis.	It is hypothesized that this pathway regulates synaptic plasticity.	Creates unnecessary distance between authors and claim.
Our findings support the use of early intervention strategies.	Strong and straightforward interpretation.	The use of early intervention strategies is supported by our findings.	Longer and less natural.
We conclude that the treatment is safe and well tolerated.	Clear ownership of the conclusion.	It is concluded that the treatment is safe and well tolerated.	Awkward and unnecessarily formal.
Table 2 shows a significant increase in gene expression.	Directly points the reader to the evidence.	A significant increase in gene expression is shown in Table 2.	Less concise and harder to read.
Figure 3 illustrates the relationship between age and disease severity.	Makes the figure the active subject.	The relationship between age and disease severity is illustrated in Figure 3.	Longer and less direct.

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1. Voice: Active Voice is preferred over Passive Voice

- Passive voice could be acceptable when:

describing procedures, materials, or events where the actor is obvious or unimportant → mostly in the methods section

Example	Passive Voice	Active Voice	Why Passive Voice May Be Better
Clinical event	The patient was transferred to the ICU.	The clinical team transferred the patient to the ICU.	The important information is the patient's transfer and current status, not who performed the transfer.
Laboratory procedure	Blood samples were centrifuged at 3,000 × g for 10 minutes.	We centrifuged blood samples at 3,000 × g for 10 minutes.	The focus is on the sample processing procedure, not on the person operating the centrifuge.

- Some journals accept passive voice (e.g. JAMA)

Always read the guidelines before you start writing!

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1. Voice: Active Voice is preferred over Passive Voice

Beyond voice: When Verbs *Describe*, Not Act

The verb **to be** is often used as a **linking (copula)** verb to connect the subject to a description, classification, property, or result – without any "action".

Examples:

- The sample **was** contaminated with bacterial DNA.
- Gene expression levels **were** significantly higher in treated cells.
- The protein **is** a key regulator of cell division.
- The observed phenotype **was** consistent with previous findings.
- These results **are** indicative of an immune response.
- The experimental group **was** more resistant to infection than the control group.

These are all typical of the **concise, objective style** used in life science theses and journal articles.

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2. Tenses: Simple tenses (present and past) are preferred over perfect tense

In scientific writing, the **simple past** and **simple present** are the preferred verb tenses because they distinguish between:

- completed actions
- unpublished/new versus established knowledge

Think of the function the sentence is conveying!

- The **simple past** is used for:

unpublished or new knowledge or **completed actions**, including the methods and results of a specific study.
(*e.g., "We measured…";"Participants completed a questionnaire."; "We found that…"*).

- The **simple present** is used for:

established knowledge, including generally accepted facts and **interpretations** that are considered true beyond a single study.
(*e.g., "Protein X regulates cell growth"; "These finds suggest a protective effect"*).

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2. Tenses: Simple tenses (present and past) are preferred over perfect tense

Very simple non scientific examples to think of the ”function”:

Simple past		*Simple present*
Did you brush your teeth? Yes, I **brushed** my teeth. (implies a completed task or new knowledge)	Vs.	Do you brush your teeth? Yes, I **brush** my teeth. (implies an established habit)
I **met** my friends. (implies a completed task or new knowledge)	Vs.	I **meet** my friends. (implies an established habit)
Her mom **baked** cookies. (implies a completed task or new knowledge)	Vs.	Her mom **bakes** cookies. (implies an established habit)

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2. Tenses: Simple tenses (present and past) are preferred over perfect tense

A useful example:

(A) We found that hypoxia **increased** VEGF expression. *(simple past)* (**new knowledge**; Results)

(B) Hypoxia increases VEGF expression. *(simple present)*
(**established knowledge**; conclusion/discussion; followed by several citations)

(C) Our results suggest that hypoxia **increases** VEGF expression. *(simple present)*
(**interpretation**; conclusion/discussion)

This transition from **results** to **interpretation** is one of the most important tense distinction in scientific writing.

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2. Tenses: Simple tenses (present and past) are preferred over perfect tense

Although **simple tenses** are preferred, **perfect tenses** can be useful in specific situations.

- It is formed with: **a form of "have" (have, has, had) + the past participle of the main verb**
  (e.g., *has studied*, *have written*, *had completed*)

- The **present perfect** is often used to describe previous research with ongoing relevance (followed by citations):

Present perfect: *"Researchers have identified several pathways involved in this process."*
"Previous studies have reported similar findings in related species."
"Several studies have shown that protein X is involved in cell growth."

- The **past perfect** can clarify the sequence of past events (usually in the methods):

Past perfect: *"The reaction had reached equilibrium before the enzyme was added."*
"The participants had completed the questionnaire before the intervention began."
"Samples had been stored for 24 h before analysis".

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2. Tenses: Simple tenses (present and past) are preferred over perfect tense

Purpose/Function conveyed	Typical tense	Section that ***usually*** contains
Established knowledge	Simple Present	Abstract, Introduction, conclusion, discussion
Previous studies	Simple Past or Present Perfect	Abstract, Introduction, conclusion, discussion
Completed actions	Simple Past	Abstract, Methods used in this study
Unpublished knowledge	Simple Past	multiple (abstract, results, conclusion, discussion)
Interpretation of results	Simple Present	Conclusion, discussion
Sequence of past events	Past Perfect	multiple

Mention of Table and figures are given in the present tense, as they provide a simple description.
e.g. Figure 1 illustrates…..; Table 2 shows….

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3. Fewer prepositions: Avoid excessive prepositional phrases

- **Prepositions** express a relationship between words, usually noun or pronoun and another element in a sentence.
  For example: *of, in, on, by, for, with,* and *to* (e.g., "analysis of data", "effect on growth").

- Use prepositions when they **add meaning**.
  They are essential for expressing relationships accurately and unambiguously in scientific writing.

- BUT **avoid unnecessary preposition chains**.
  Long strings such as "the analysis **of** the effect **of** the treatment **on** the growth **of** the plants" can often be simplified.

- Prefer **strong nouns and verbs** over prepositional phrases.
  Instead of "performed an analysis of the data," write "analyzed the data." This makes sentences shorter and more direct.

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3. Fewer prepositions: Avoid excessive prepositional phrases

Prepositions are essential for precision, but excessive use can make sentences unnecessarily long and indirect.

Wordy: "The investigation **of** the effect **of** temperature **on** the rate **of** reaction was conducted."

Better: "We investigated how temperature affected the reaction rate."

The most frequently overused prepositions are **of, in, for, on, by,** and **with.**
They often appear in long noun phrases (e.g., "the analysis of the effect of…"),
which can usually be rewritten more directly.

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3. Fewer prepositions: Avoid excessive prepositional phrases

Type of Preposition	Purpose	Examples
Place	Indicates where something is located	in, on, at, under, over, between, among, beside, behind, near, inside, outside
Time	Indicates when something occurs	at, on, in, before, after, during, since, until, by, within, throughout
Movement/Direction	Indicates movement from one place to another	to, toward, into, onto, from, through, across, along, around, past, out of

Examples:
*The samples were stored **in** a refrigerated chamber. (Place)*
*Measurements were recorded **during** the experiment. (Time)*
*The solution was transferred **into** a sterile container. (Movement)*

Type of Preposition	Purpose	Examples
Limitation	Indicates exclusion, exceptions	but, except, without
Location / Spatial Relationship	Indicates position relative to another object	above, below, under, over, beside, between
Agent / Instrument	Indicates who performs an action or the means by which it is performed	by, with, through, via

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3. Fewer prepositions: Avoid excessive prepositional phrases

Tips to reduce prepositions	Example
1. Replace noun–preposition combinations with strong verbs.	Wordy: We conducted an analysis of the data. Better: We analyzed the data.
2. Reduce long chains of "of" phrases.	Wordy: The analysis of the effect of the treatment on the growth of the plants... Better: The analysis of treatment effects on plant growth...
3. Convert prepositional phrases into adjectives.	Wordy: The cells in the liver were examined. Better: The liver cells were examined.
4. Use possessives instead of "of" constructions when appropriate.	Wordy: The results of the experiment were significant. Better: The experiment's results were significant.
5. Remove redundant prepositional phrases.	Wordy: The samples were collected at a time before analysis. Better: The samples were collected before analysis.
6. Rewrite sentences to make the subject perform the action directly (active voice)	Wordy: A measurement of temperature was performed by the researchers. Better: The researchers measured the temperature.

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4. Fewer modifiers: keep clear and concise

Modifiers are words or phrases that add detail to another word, making its meaning more specific or precise.

Why use fewer modifiers?
They improve precision, but excessive use often makes scientific writing **wordy, subjective, and less direct**.

Class of modifier	Purpose	Examples
Adjectives	Modify nouns	**significant** effect, **rapid** growth, **cellular** response
Adverbs	Modify verbs, adjectives, or other adverbs	**significantly** increased, **highly** expressed, **strongly** correlated
Nouns (attributive nouns)	Modify other nouns	**protein** expression, **cell** culture medium, **gene** regulation pathway

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4. Fewer modifiers: keep clear and concise

Note that **Noun modifiers** are often more concise than adjective or prepositional constructions.

the concentration of the protein *vs* *the protein concentration*

the regulation of genes *vs* *gene regulation*

BUT long strings of nouns quickly become difficult to read.

cell culture medium protein expression analysis protocol vs Protocol for protein expression analysis in cell culture media

Balance brevity with clarity.

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4. Fewer modifiers: keep clear and concise

Tip	Example
1. Remove empty intensifiers	**Very** large increase → large increase or, better, a 45% increase
2. Replace modifiers with data	**highly** expressed gene → gene expression increased 3.2-fold
3. Keep only modifiers that add essential information	**carefully** measured protein concentration → protein concentration was measured
4. Avoid stacking multiple modifiers on the same word	a **highly** significant and **remarkably rapid** increase → a significant increase

Be aware: adverbial modifiers can make a sentence sound more subjective rather than quantitative.

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5. Punctuation: Clear marks, clear meaning

Punctuation marks are symbols that organize sentences and clarify relationships between ideas.

- Correct punctuation improves **clarity** and **prevents ambiguity**.
- It helps readers follow complex scientific information **efficiently**.
- Misused punctuation can change meaning, create confusion, or obscure important results.
- Consistent punctuation contributes to a professional and credible scientific style.

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5. Punctuation: Clear marks, clear meaning

Example of how punctuation can subtly change meaning and interpretation:

I enjoy cooking, my colleagues, and my dog. (a list of things that I enjoy)	vs	I enjoy cooking my colleagues and my dog. (what I enjoy cooking for dinner)
The samples that were contaminated were discarded. (Only the contaminated samples were discarded.)	vs	**The samples, which were contaminated, were discarded.** (All samples were contaminated and all were discarded.)

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5. Punctuation: Clear marks, clear meaning

Punctuation mark	Main use	Example
Comma (,)	After an introductory word (often and adverb) in a sentence	However, the difference was not statistically significant
	To set off a nonessential clause	Scientists, who often work overtime, tend to lack time for writing manuscripts.
	After an introductory (prepositional or adverbial) phrase	Generally speaking, this is sound advice
	To separate two independent clauses in a compound sentence	All blood samples were processed, and IgG levels determined.
	To set off appositives or interrupting words	Alzheimer's disease, named after Dr. Alois Alzheimer, is the most common form of dementia.
Serial (Oxford) comma (,)	The final comma before 'and' or 'or' in a list of three or more items; helps avoid ambiguity and improves clarity in complex lists.	The experiment measured temperature, pH, and cell density.

Source (based on): Mastering scientific and medical writing – A self-help guide; 2nd Edition, Silvia M. Rogers; Appendix 11.3

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5. Punctuation: Clear marks, clear meaning

Punctuation mark	Main use	Example
Semicolon (;)	Connects closely related independent clauses or separates complex list items that already contain commas. Easily replaced by a period.	The control group showed no change; the treatment group showed increased expression.
Colon (:)	Introduces a list, explanation, definition, result, or conclusion that follows from the preceding statement.	Three variables were measured: temperature, pH, and cell density.
Parentheses ( )	Enclose supplementary information, abbreviations, references, or clarifications that are not essential to the main sentence; also enclose mathematical equations	Samples were stored at −80 °C (except for the pilot study samples).
Brackets [ ]	Used for references and to enclose information added to quoted material, such as clarifications, corrections, or explanatory notes; also enclose mathematical equations	The authors stated that "[gene expression] increased significantly." [3]

Source (based on): Mastering scientific and medical writing – A self-help guide; 2nd Edition, Silvia M. Rogers; Appendix 11.3

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5. Punctuation: Clear marks, clear meaning

Punctuation mark	Main use	Example
Hyphen (-)	To link two words jointly modifying a third, unless the first of these is an adverb ending in 'ly'.	A pale-yellow granular substance; a single-center, double-blind clinical study; low-quality raw material; freshly collected samples
	To avoid confusion	The dentist's chair was re-covered.
	Use with number modifiers	A 20-year old subject; a two-year study
	if a vowel ends the prefix and begins the root word, especially if the vowel is repeated.	anti-inflammatory, pre-existing, follow-up
	Hyphens are no longer necessary for intra, inter, pre, post, non, re, sub, etc.	intrasubject, nonessential, postdosing, recalculated, preclinical, anticancer
Dash (—)	Adds emphasis or an explanatory aside; generally used sparingly in scientific writing because it can disrupt sentence flow. Avoid!	The result was unexpected—the protein level increased rather than decreased.

Source (based on): Mastering scientific and medical writing – A self-help guide; 2nd Edition, Silvia M. Rogers; Appendix 11.3

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5. Punctuation: Clear marks, clear meaning

Punctuation mark	Main use	Example
Quotation marks (" ")	Indicate direct quotations or highlight specific terms, labels, or wording taken from another source.	The authors described this phenomenon as "rapid adaptation."
Apostrophe (')	Indicates possession; contractions are generally avoided in formal scientific writing.	The patient's response was recorded.
Exclamation mark (!)	Expresses strong emotion, surprise, or emphasis; rarely appropriate in scientific writing because it can appear subjective or informal. Just don't use it!	Avoid: "The treatment produced remarkable results!"

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WRITING A CLEAR SCIENTIFIC MESSAGE
Style rules

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1. Narrative thread: keeping the scientific story clear and coherent
2. Sentence Design: concise sentences and logical and clean syntax
3. Spelling: Following American or British English consistently
4. Numerical style: Presenting numbers conventionally and consistently
5. Capital letters: Applying correctly and sparingly
6. Abbreviations: Define and use consistently

Note 4 main words: clear, concise, convention and consistency

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1. Narrative thread: keeping the scientific story clear and coherent

- Scientific writing follows the same core grammar rules as standard English, but it applies them with a strong emphasis on **clarity, precision, objectivity and consistency.**

- Scientific storytelling is not about being dramatic or sounding complex, it's about helping the reader understand the significance and logical flow of the work, clearly, without ambiguity.

- Main tip: **Know your data before you start writing, so you can tell a clear story**.

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1. Narrative thread: keeping the scientific story clear and coherent

Before writing, make sure you can clearly explain your question, methods, results, and conclusions.

A strong paper or thesis simply tells this scientific story with logical, clear and coherent sequences.

Step	Guiding Question	Brief Explanation
Research Question	What did we want to find out?	Define the problem, knowledge gap, or hypothesis that motivated the study.
Methods	How did we test it?	Describe the experimental design, materials, and procedures used to answer the question.
Results	What did we find?	Present the observations and data obtained from the study.
Conclusions	What do the findings mean?	Interpret the results and explain how they answer the original question.

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2. Sentence Design: concise sentences and logical and clean syntax

Syntax: The way words are organized in a sentence to create clear and grammatically correct meaning.

- **Improves readability** – Shorter sentences are easier to read and understand on the first pass.
- **Reduces ambiguity** – Clear syntax helps readers interpret your meaning correctly and avoid misunderstandings.
- **Highlights key findings** – Simple sentence structure keeps attention on the scientific message rather than the wording.

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2. Sentence Design: concise sentences and logical and clean syntax

Syntax and subject-verb agreement → the verb must match the subject in **number** (singular or plural).

- **Singular subjects take singular verbs; plural subjects take plural verbs.**
  *The experiment **shows**…* vs. *The experiments **show**…*

- **Identify the true subject before choosing the verb.**
  Additional phrases between the subject and verb do not change agreement.
  *The results of the study **show**…*

- **Collective nouns are usually singular in scientific writing.**
  *The research team/the committee/ **was**…*

- **Words such as "each," "every," and "either/neither" take singular verbs.**
  *Each sample **was** analyzed.*

- **Compound subjects joined by "and" usually take a plural verb.**
  *The treatment and control groups **were** compared.*

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2. Sentence Design: concise sentences and logical and clean syntax

Rule of Meaning (Notional Agreement): Sometimes the verb follows the idea being expressed rather than the grammatical form of the subject. In scientific writing, when the emphasis is on the individuals within a group (patients, participants, genes, cells), a plural verb is often clearer and more natural.

Grammatically Correct but Less Meaningful	Preferred by Rule of Meaning
A number of cells was observed in the sample.	A number of cells were observed in the sample.
A majority of participants was satisfied with the treatment.	A majority of participants were satisfied with the treatment.
A total of 128 genes was identified in the analysis.	A total of 128 genes were identified in the analysis.

- A total of 41 patients **were** enrolled in the study. ✓ (focus on the patients)
- The total number of patients **was** 41. ✓ (focus on the total itself)

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2. Sentence Design: concise sentences and logical and clean syntax

Tricky examples when singular verb is wrong:

Incorrect	Correct	Why?
The causative bacteria **was** determined by culture.	The causative bacteria **were** determined by culture.	Bacteria is plural (singular: bacterium).
The data **was** analyzed descriptively.	The data **were** analyzed descriptively.	Traditionally, data is the plural of datum in scientific writing.
Several samples **was** excluded due to contamination.	Several samples **were** excluded due to contamination.	Samples is a plural noun.
These findings **suggests** a role for inflammation.	These findings **suggest** a role for inflammation.	Findings is plural.
The results of the experiment **shows** a significant difference.	The results of the experiment **show** a significant difference.	The true subject is results (plural), not experiment (singular).
The culture media **was** prepared according to the manufacturer's instructions.	The culture media **were** prepared according to the manufacturer's instructions.	*Media* is the plural of *medium*; therefore, it takes a plural verb.

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2. Sentence Design: concise sentences and logical and clean syntax

Tricky examples when plural verb is wrong:

Incorrect	Correct	Why?
The experiment **show** a significant effect.	The experiment **shows** a significant effect.	Experiment is a singular noun.
Each sample **were** analyzed separately.	Each sample **was** analyzed separately.	Each takes a singular verb.
The research team **were** responsible for data collection.	The research team **was** responsible for data collection.	In scientific writing, collective nouns such as team are usually treated as singular.
The culture medium **were** supplemented with antibiotics.	The culture medium **was** supplemented with antibiotics.	Medium is singular (plural: media).
The information obtained from the questionnaire **indicate** a need for further study.	The information obtained from the questionnaire **indicates** a need for further study.	Information is an uncountable singular noun.

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2. Sentence Design: concise sentences and logical and clean syntax

When a quantity, measurement, percentage, amount, distance, volume, or duration is treated as a single unit, it usually takes a singular verb.

Incorrect	Correct	Why?
A total of 25 mL of buffer **were** added to each sample.	A total of 25 mL of buffer **was** added to each sample.	The volume is treated as a single quantity.
Approximately 2.5 kg of tissue **were** collected during the study.	Approximately 2.5 kg of tissue **was** collected during the study.	The mass is considered one amount.
Nearly 70% of the administered dose **were** eliminated within 24 h.	Nearly 70% of the administered dose **was** eliminated within 24 h.	The percentage refers to the singular noun dose.
During the experiment, 15 L of culture medium **were** prepared.	During the experiment, 15 L of culture medium **was** prepared.	The volume functions as a single unit.
An average of 8 h of sleep **were** recorded per night.	An average of 8 h of sleep **was** recorded per night.	The duration is treated as one amount of time.

70% of the administered dose was eliminated. ✓ (*dose* = singular)
70% of the patients were discharged. ✓ (*patients* = plural)

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2. Sentence Design: concise sentences and logical and clean syntax

Syntax and **dangling participle:** They occur when a participial phrase does not clearly refer to the subject of the main clause, making the sentence confusing or unintentionally illogical.

Incorrect (Dangling Participle)	Correct
After analyzing the data, the results showed a significant difference.	After analyzing the data, we found a significant difference in the results.
This sentence implies that **the results analyzed the data**, which is not what the writer intends. The writer (or researchers) analyzed the data. This makes dangling participles a common source of unintended ambiguity in scientific papers and theses.	
Using fluorescence microscopy, several cells were identified.	Using fluorescence microscopy, the researchers identified several cells.
While reviewing the literature, several knowledge gaps were identified.	While reviewing the literature, the authors identified several knowledge gaps.

Tip: ask the question "Who is performing the action in the opening phrase?"
If the answer is not the subject that immediately follows, the participle is probably dangling.

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2. Sentence Design: concise sentences and logical and clean syntax

Syntax and **and the use of "respectively":**

Respectively indicates that two or more items correspond to another set of items in the same order.

Key Point	Example
Use only when matching items in order.	The mean ages of the control and treatment groups were 45 and 52 years, respectively.
Place it at the end of the sentence or clause.	Protein A and Protein B increased by 20% and 35%, respectively.
Avoid it when the relationship is already obvious.	The control group was 45 years old and the treatment group was 52 years old, respectively. (unnecessary)

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3. Spelling: Following American or British English consistently

Very simple:

→ Check the guidelines (journal, funding agency, uni etc):

If you have the choice, choose one and **be consistent**

If you don't, follow the convention and **be consistent**

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4. Numerical style: Presenting numbers conventionally and consistently

Why it matters: Consistent use of numbers improves readability and prevents confusion.

Guideline	Example
Use numerals for measurements, data and statistics	5 mL, 12 patients, 37°C, 8.4%, p = 0.03
Spell out small numbers used in general text (typically one to nine) unless a *journal style guide specifies otherwise*.	Three samples failed quality control.
Use numerals when a sentence contains a mix of small and large numbers	The study included 8 mice and 15 rats.
Do not begin a sentence with a numeral	Incorrect: 25 participants completed the survey. Correct: Twenty-five participants completed the survey.
Use the same format for comparable values	Consistent: 5, 8, and 12 samples Inconsistent: five, 8, and twelve samples
Always follow journal, university, or style-guide requirements	Specific rules may vary between publications.

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4. Numerical style: Presenting numbers conventionally and consistently

Some examples:

Incorrect	Correct	Explanation
The study included five mice, 8 rats, and twelve rabbits.	The study included **5 mice, 8 rats, and 12 rabbits**.	Use a consistent format for comparable values.
12 samples were excluded from the final analysis.	**Twelve samples** were excluded from the final analysis.	Avoid starting a sentence with a numeral.
The cells were incubated for **five hours** at 37°C.	The cells were incubated for **5 hours** at 37°C.	Use numerals for measurements and experimental conditions.
The treatment group contained **15 subjects**, while the control group contained **fifteen subjects**.	The treatment group contained **15 subjects**, while the control group contained **15 subjects**.	Use the same format for comparable values within a sentence.

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4. Numerical style: Presenting numbers conventionally and consistently

key message: **consistency and adherence to the target journal's style guide**

Feature	European Convention	British/American Convention	Recommended for International Scientific Writing
Decimal separator	Comma (,)	Period (.)	**Period (.)**
Example	3,14	3.14	**3.14**
Thousands separator	Period (.), space, or apostrophe (') depending on country	Comma (,)	**Thin space or regular space** (SI recommendation)
Example	1.234 or 1'234 or 1 234	1,234	**1 234**
Large numbers	1.234.567 or 1 234 567	1,234,567	**1 234 567**
Percentages	25,5%	25.5%	**25.5%**
Scientific measurements	5,7 mL	5.7 mL	**5.7 mL**
Use in international journals	Rare	Common	**Preferred**

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5. Capital letters: Applying correctly and sparingly

Use capital letters only when required

- Capitalize **proper nouns** (specific names of people, places, organizations, software, and trademarks).
- Capitalize the **first word** of a sentence, heading, or title (according to the style guide being used).
- Do **not** capitalize common nouns simply because they seem important or technical.
- If you chose capitals in titles, do not capitalize articles, conjunctions, and prepositions.
  (*e.g.:* "**I**ncomplete **R**emoval of **E**xtracellular **G**lutamate **C**ontrols **S**ynaptic **T**ransmission and **I**ntegration at a **C**erebellar **S**ynapse"
- Be **consistent** with capitalization throughout the document.

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5. Capital letters: Applying correctly and sparingly

Common Pitfalls

- **Alzheimer disease**, **Parkinson disease**, **Gram stain** (named after people → capitalize the name)
- **type 2 diabetes**, **breast cancer**, **hypertension, dementia** (common disease names → lowercase)
- **Figure 1**, **Table 2**, **Section 3** (when referring to a specific numbered item)
- **the figure**, **the table**, **the section** (general reference → lowercase)

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5. Capital letters: Applying correctly and sparingly

Incorrect	Correct	Explanation
We analyzed samples from the **university of basel**.	We analyzed samples from the **University of Basel**.	Official names of institutions are proper nouns and should be capitalized.
The study investigated **Diabetes Mellitus**.	The study investigated **diabetes mellitus**.	Most disease names are common nouns and should not be capitalized unless they contain a proper name.
We used **Western Blotting** and **Polymerase Chain Reaction**.	We used **western blotting** and **polymerase chain reaction**.	Scientific methods and techniques are generally common nouns and are written in lowercase.
The samples were analyzed using **microsoft excel**.	The samples were analyzed using **Microsoft Excel**.	Product, software, and trademark names follow the capitalization chosen by the manufacturer.
As shown in **figure 2**, the treatment increased survival.	As shown in **Figure 2**, the treatment increased survival.	Capitalize Figure, Table, Appendix, Chapter, or Section when referring to a specific numbered item.
The cells were stained using the **gram stain** method.	The cells were stained using the **Gram stain** method.	Terms derived from a person's name retain capitalization of the proper name.
The Patients were diagnosed with **Type 2 Diabetes** and treated at the **University hospital Basel**.	The patients were diagnosed with **type 2 diabetes** and treated at the **University Hospital Basel**.	Common disease names are lowercase, whereas official institution names are capitalized.

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6. Abbreviations: Define and use consistently

Why use abbreviations?

- Reduce repetition of long terms and improve readability (less wordy).
- Use only when a term appears multiple times and the abbreviation is widely recognized or clearly useful.
- **If readers must frequently stop to remember what an abbreviation means, you are using too many abbreviations.** Abbreviations should improve readability to communicate the science, not create extra work for the reader.

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6. Abbreviations: Define and use consistently

Key Rules

- Define abbreviations the first time they appear:
  Reactive oxygen species (ROS) contribute to oxidative stress.
- Use the abbreviation consistently after definition.
- Avoid introducing too many abbreviations. Readers should not need a glossary to follow your paper.
- Do not alternate between the full term and abbreviation unless clarity requires it.

Example

Bad: Reactive oxygen species (ROS) were measured. Reactive oxygen species increased after treatment. ROS levels remained elevated.

Good: Reactive oxygen species (ROS) were measured. ROS levels increased after treatment and remained elevated.

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6. Abbreviations: Define and use consistently

Type	Examples	How to Use
Standard scientific abbreviations	DNA, RNA, ATP, PCR, WHO	Usually do not need definition because they are universally recognized in the field. Follow journal guidelines.
Study-specific abbreviations	Reactive oxygen species (ROS), High-fat diet (HFD)	Define at first mention, then use the abbreviation consistently throughout the document.
Units and measurements	mg, mL, °C, kg	Use standard SI abbreviations; do not define them.
Statistical abbreviations	SD, SEM, CI, ANOVA	Define if there is any possibility of ambiguity, especially for student theses.

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Conclusion: Grammar and Style Serve the Science

- **Good scientific writing is more than correct grammar.** It guides readers through a clear, logical, and convincing scientific story.
- **Grammar and style choices shape understanding.** Appropriate use of tense, voice, punctuation, numbers, abbreviations, and sentence structure reduces ambiguity and improves readability.
- **Every sentence should have a purpose.** Present information in a way that helps readers follow the progression from question → methods → results → conclusions.
- **Think beyond rules.** Effective scientific writers understand the *function* of each section and choose the language that best communicates their **scientific message**.

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Remember:

Readers should focus on your science, not struggle with your writing.

Clear grammar and consistent style make the story of your research easier to understand, evaluate, and remember.

Final Take-Home Message

Scientific writing is not just about reporting data.
It is about telling a clear, credible, and coherent scientific story.